MALLA REDDY UNIVERSITY

An Application Development Project Report

DESIGN AND IMPLEMENTATION OF A WIRELESS ESP32 ROBO WAR

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We hereby declare that the project report **"DESIGN AND IMPLEMENTATION OF A WIRELESS ESP32 ROBO WAR",** has been carried out by us and this work has been submitted to the **Department of Electronics and Communication Engineering (ECE), Malla Reddy University, Hyderabad**. We further declare that this project work has not been submitted in full or part for the award of any other degree in any other educational institutions.

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ABSTRACT

The Robo War is an ESP32-based combat robot developed for controlled robotic competitions. The main aim of the project is to design a compact, strong, reliable, and easily controllable robot capable of moving in forward, backward, left, and right directions. The system combines electronic control, motor driving, power management, and mechanical construction to create a functional robotic platform.The ESP32 microcontroller is used as the main control unit of the Robot War. It receives commands from the wireless control system and processes them to control the motors. The robot is programmed using Arduino IDE with Embedded C/C++. The ESP32 provides the required control logic for coordinating the movement of the robot.

The robo uses 300 RPM DC motors and BTS7960 motor drivers for controlling motor speed and direction. The motor-control system allows the robot to perform forward, backward, left, and right movements. Four wheels are used for movement and stability, while the chassis supports and protects the electronic and mechanical components.A 12V 7Ah battery is used as the main power source for the robot. An LM2596 voltage regulator provides a suitable voltage for the ESP32 and other electronic components. Proper power distribution helps maintain stable operation of the controller and motor-driving system during robot operation.
The development process includes mechanical assembly, electrical connections, ESP32 programming, motor-control implementation, wireless control, and system testing. The robot is tested for movement, motor performance, speed, stability, and wireless-control response. These tests help verify the proper functioning of the complete robotic system.

The project also provides practical experience in embedded systems, robotics, motor control, wireless communication, and power management. The integration of the ESP32, motor drivers, DC motors, battery, regulator, wheels, and chassis demonstrates the working of a complete robotic system.Overall, the Robot War provides a simple and practical platform for understanding robotic control and embedded system applications. The developed prototype can be further improved by adding stronger motors, additional sensors, improved chassis construction, and other mechanisms to enhance its performance and capabilities.

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CHAPTER - 1

INTRODUCTION

Robotics is an important field of engineering that combines electronics, programming, mechanical design, and control systems to develop machines capable of performing different tasks. Mobile robots are widely used for applications such as automation, inspection, competition, and research. A Robot War is a mobile robotic system designed to operate in a controlled arena and compete against another robot while maintaining proper movement and stability.The Robot War developed in this project uses an ESP32 microcontroller as its main control unit. The ESP32 receives movement commands from a wireless control system and processes them to control the robot motors. Its control capability and wireless communication features make it suitable for developing a compact and easily controllable robotic platform.

The robot uses 300 RPM DC motors along with BTS7960 motor drivers for controlling motor speed and direction. The motor drivers receive control signals from the ESP32 and operate the motors according to the required movement. This arrangement allows the robot to move in forward, backward, left, and right directions.The power system consists of a 12V 7Ah battery and an LM2596 voltage regulator. The battery provides power to the motors, while the voltage regulator supplies a suitable voltage to the ESP32 and other electronic components. A four-wheel arrangement and a strong chassis are used to provide proper support and stability during operation.

The project focuses on developing a simple, reliable, and practical Robot War using easily available components. The development includes hardware assembly, electrical connections, programming, wireless control, motor testing, and overall system testing. The robot is evaluated based on movement, motor performance, speed, stability, and control response.The developed Robot War provides practical knowledge of ESP32 programming, DC motor control, motor drivers, wireless communication, power management, and robotic system design. The project also provides a basic platform that can be further improved by adding stronger motors, sensors, improved mechanical structures, and additional mechanisms for enhanced robotic performance.

1.1 Problem Definition & Description

The main problem addressed in this project is the development of a strong, compact, and easily controllable Robot War capable of operating in a controlled competition arena. The robot must be able to move smoothly in forward, backward, left, and right directions while maintaining proper stability during operation.The system requires reliable control of DC motors for achieving the desired movement and direction. Therefore, an ESP32 microcontroller is used as the main controller, along with BTS7960 motor drivers to control the speed and direction of the motors. The robot also requires a suitable power supply to operate the motors and electronic components effectively.

Another important challenge is to provide sufficient mechanical strength to withstand impacts during competition. The chassis, wheels, motors, battery, and electronic components must be properly assembled to maintain stability and reliable operation.The project aims to solve these challenges by integrating the ESP32, DC motors, BTS7960 motor drivers, 12V battery, LM2596 regulator, wheels, and chassis into a single robotic system. The completed Robot War is tested for movement, speed, stability, motor performance, and wireless control to ensure that it meets the required operational objectives.

1.2 Aim and Objectives of the Project

The main aim of this project is to design and develop a strong, compact, low-cost, reliable, and easily controllable Robot War using an ESP32 microcontroller, 300 RPM DC motors, BTS7960 motor drivers, a 12V 7Ah battery, and an LM2596 voltage regulator. The robot is designed to achieve smooth and controlled movement in forward, backward, left, and right directions while maintaining proper stability during operation. The project focuses on integrating the hardware and software components into a single functional robotic system and implementing wireless control for convenient operation. The objectives include designing a strong robot chassis, connecting and controlling the DC motors through BTS7960 drivers, providing a suitable power supply for the controller and motors, programming the ESP32 for movement control, and testing the robot for speed, stability, motor performance, and wireless response. Through this project, practical knowledge of embedded programming, motor control, power management, wireless communication, and robotic system development is gained. The final objective is to develop a fully working Robot War prototype that can operate effectively in a controlled competition environment and provide a base for future improvements and advanced robotic features.

1.3 SCOPE OF THE PROJECT

The scope of this project is to design and develop an ESP32-based Robot War capable of reliable movement and wireless control in a controlled competition environment. The project covers the development of the mechanical structure, motor-control system, power supply, wireless communication, programming, testing, and performance evaluation of the robot.

1. ESP32-Based Control System

The project includes the use of an ESP32 microcontroller as the main control unit of the Robot War. The ESP32 receives movement commands and processes them to generate the required control signals for operating the motors.

2. DC Motor Control

The project includes the integration of 300 RPM DC motors for driving the robot. The motors are controlled using BTS7960 motor drivers, which provide control over the speed and direction required for different robot movements.

3. Directional Movement

The scope includes implementing the basic movements required for operating the Robot War. The robot is designed to move smoothly in forward, backward, left, and right directions by controlling the motors according to the received commands.

4. Wireless Control

Wireless communication is included to provide remote operation of the Robot War. The operator can send movement commands to the robot, and the ESP32 processes these commands to control the motor drivers and achieve the required movement.

5. Power Supply System

The project includes a 12V 7Ah battery as the main power source for the robot. An LM2596 voltage regulator is used to reduce the battery voltage to a suitable level for the ESP32 and other electronic components, ensuring stable operation of the control system.

6. Mechanical Structure

The scope includes the design and assembly of the robot chassis, wheels, motors, and other mechanical components. The chassis is intended to provide sufficient support and stability while also protecting the electronic components during robot operation and controlled battle conditions.

7. Motor and System Integration

The project involves integrating the ESP32, BTS7960 motor drivers, DC motors, battery, voltage regulator, wheels, and chassis into one complete robotic system. Proper electrical and mechanical connections are required to ensure coordinated operation of all major components.

8. Testing and Performance Evaluation

The developed Robot War is tested for forward, backward, left, and right movement along with speed, stability, motor performance, and wireless-control response. The testing process helps identify connection, programming, and mechanical issues and confirms the overall functionality of the robot.

9. Future Enhancement

The developed Robot War provides a foundation for further development and improvement. Future versions can include stronger motors, additional sensors, improved chassis construction, and a weapon mechanism to increase the robot's performance and capabilities in advanced competition environments.

CHAPTER – 2

LITERATURE AND EXISTING SYSTEM

2.1 Existing System Analysis:

Existing Robot Wars and small combat robots generally use DC motors, motor drivers, batteries, and remote-control systems to achieve basic robotic movement. These systems are designed to move the robot in different directions such as forward, backward, left, and right. Conventional systems commonly use wired controllers, RF modules, Bluetooth, or other wireless communication methods for sending movement commands to the robot. The overall performance of these systems mainly depends on the motor capacity, motor driver, power supply, communication range, and mechanical strength of the robot chassis.In many existing systems, DC motors are controlled using motor driver modules that provide the required current for the motors. However, basic motor control systems may have limitations in terms of response speed, control flexibility, and smooth directional movement. Some systems also use low-current motor drivers that may not be suitable for high-power motors used in Robot Wars. Therefore, a suitable high-current motor driver is required to provide reliable motor operation and better control during movement.

The existing systems also face challenges related to power consumption and power management. Motors generally require a higher voltage and current compared with the microcontroller and other electronic components. Improper power distribution can cause voltage drops, unstable controller operation, or reduced motor performance. A suitable battery and voltage regulation system are therefore important for maintaining stable operation of both the motors and control electronics.

Mechanical construction is another important limitation in conventional robotic systems. A weak or lightweight chassis may not withstand impacts and vibrations during operation. Proper placement of motors, wheels, battery, motor drivers, and other components is required to achieve good weight distribution and stability. The robot should also be compact enough to provide effective movement

2.1.1 Background & Literature Survey

Battle robots are a practical application of robotics, embedded systems, motor control, and wireless communication. The development of mobile robotic systems has focused on achieving reliable movement, accurate control, efficient power utilization, and strong mechanical construction. In combat-oriented robots, these requirements become more important because the robot must maintain stability and respond quickly to movement commands while operating under mechanical stress.Earlier mobile robotic systems commonly used DC motors along with motor driver circuits to control the speed and direction of the wheels. Different motor driver modules have been developed to handle the higher current requirements of DC motors while allowing a microcontroller to provide low-power control signals. The selection of an appropriate motor driver is important because the controller itself cannot directly supply the current required by high-power motors.

Microcontrollers have played an important role in the development of modern robotic systems. Traditional robotic platforms used different controller boards for motor control and communication, while newer systems increasingly use microcontrollers with built-in wireless communication capabilities. The ESP32 is suitable for such applications because it combines processing capability with wireless communication features, allowing it to receive commands and control external hardware through programmed control logic.Wireless control has also become an important feature in mobile and Robot War systems. RF, Bluetooth, and Wi-Fi-based approaches can be used to transmit commands from a remote controller to the robot. Wireless operation eliminates the need for physical control cables and provides greater flexibility during testing and operation. The effectiveness of a wireless robotic system depends on communication reliability, response time, and the control program implemented in the microcontroller.

Power management is another major consideration identified in robotic system development. DC motors generally require higher current than the microcontroller and communication circuits. Therefore, a separate power arrangement and voltage regulation are required to supply appropriate voltage levels to different parts of the system. In the proposed Robot War, a 12V 7Ah battery is used as the main power source, while an LM2596 voltage regulator provides a suitable regulated supply for the low-voltage electronics.

2.1.2 Limitations of Existing System

Existing Robot War systems have limitations in wireless control, motor performance, power management, and mechanical stability. Basic motor drivers may not support the high current required by powerful DC motors, while improper power management can cause unstable operation. Weak chassis designs may also reduce stability during movement and impact. Limited wireless range and delayed response can affect accurate control. These limitations create the need for a stronger, reliable, and better-integrated Robot War system using an ESP32, BTS7960 motor drivers, suitable power supply, and a strong chassis.

2.2 Proposed System Analysis

The proposed system is an ESP32-based Robot War developed to provide reliable, stable, and easily controllable robotic movement. The system integrates the microcontroller, motor drivers, DC motors, power supply, chassis, and wireless control into a single robotic platform. The ESP32 acts as the main control unit and processes the commands received from the wireless controller. Based on the received commands, the ESP32 controls the direction and movement of the robot.

The robot uses 300 RPM DC motors to provide the required movement and driving force. BTS7960 motor driver modules are used to control the motors because they can handle the higher current required by the DC motors. The motor drivers receive control signals from the ESP32 and regulate the motor operation according to the selected direction. This arrangement allows the robot to move in forward, backward, left, and right directions with proper control.

For the power system, a 12V 7Ah battery is used as the main power source for the Robot War. Since the ESP32 and other electronic components require a suitable regulated voltage, an LM2596 voltage regulator is used to provide the required lower voltage supply. Proper power distribution helps maintain stable operation of the controller and motor control system during movement.

2.2.1 Proposed System Description

The proposed system is an ESP32-based Robot War developed to provide reliable wireless control, stable movement, and easy operation. The ESP32 acts as the main controller of the robot and processes the commands received from the wireless control system. Based on these commands, the controller operates the motors through the motor driver circuit and controls the required direction of movement.

The Robot War uses 300 RPM DC motors to provide the required driving force. BTS7960 motor drivers are used to control the motors and handle the higher current required during operation. The system allows the robot to move in forward, backward, left, and right directions. The motor drivers

receive control signals from the ESP32 and operate the motors according to the selected movement command.

A 12V 7Ah battery is used as the main power source for the robot. Since the ESP32 and other electronic components require a suitable regulated voltage, an LM2596 voltage regulator is used to reduce the voltage to the required level. Proper power distribution helps maintain stable operation of the controller and other electronic components.The motors, wheels, battery, ESP32, motor drivers, and other components are mounted on a suitable chassis to provide proper balance and mechanical stability. The complete system is tested for directional movement, motor performance, speed, wireless response, and stability. Thus, the proposed system integrates the hardware, software, power supply, and mechanical components into a functional Robot War prototype

2.2.2 Advantages of Proposed System

The proposed Robot War provides reliable wireless control, smooth directional movement, and stable motor operation. The ESP32 acts as the main controller and provides flexible programming and control of the robot. The BTS7960 motor drivers provide effective control of the 300 RPM DC motors, allowing the robot to move forward, backward, left, and right. The 12V 7Ah battery supplies the required power, while the LM2596 voltage regulator provides a suitable voltage for the electronic components. The system is simple to operate, easy to program, and provides good stability during movement. The design also allows future improvements such as adding sensors, stronger motors, and advanced robotic mechanisms..

2.3 Software & Hardware Requirements

The development of the proposed Robot War requires both software and hardware components. The hardware provides the physical control, movement, and power functions, while the software is used to program the ESP32 and implement the required wireless motor-control operations.

2.3.1 Software Requirements

The proposed Robot War requires suitable software tools for programming, controlling, testing, and debugging the robotic system. Arduino IDE is used as the main development environment for writing, compiling, and uploading the program to the ESP32 microcontroller. The ESP32 board package is installed in Arduino IDE to provide the required support for programming the selected controller.

The control program is developed using Embedded C/C++, which is used to implement the movement logic and control the BTS7960 motor drivers. The program processes the received wireless commands and generates the required control signals for the motors. Different commands are programmed for forward, backward, left, and right movements of the Robot War.The Serial Monitor available in Arduino IDE is used during development to monitor program execution and troubleshoot communication or control-related issues. It helps in checking the received commands and verifying the operation of the control program. The software can also be modified when required to improve motor control, wireless response, and overall robot performance.

Thus, Arduino IDE, ESP32 board support, Embedded C/C++, and Serial Monitor together provide the necessary software environment for developing, programming, testing, and debugging the ESP32-based Robot War.

2.3.2 Hardware Requirements

1. ESP32 Microcontroller

The ESP32 is the main controller of the Robot War. It is used to process wireless commands and generate control signals for the motor drivers. It operates at 3.3 V and provides built-in Wi-Fi and Bluetooth communication, making it suitable for wireless robotic control applications.

2. 300 RPM DC Motors

The 300 RPM DC motors are used to provide the required driving force for the Robot War. These geared motors provide suitable torque and speed for moving the robot. The motors are connected to the BTS7960 motor drivers, which control their direction and operation according to the commands received from the ESP32.

3. BTS7960 Motor Driver

The BTS7960 is a high-current motor driver used to control the DC motors. It allows the motors to operate in forward and reverse directions and can handle the higher current required by the motors. The driver receives control signals from the ESP32 and provides the required power to the motors.

4. 12V 7Ah Battery

The 12V 7Ah battery is the main power source of the Robot War. It provides the required electrical power for the DC motors and other system components. The battery capacity of 7 Ah allows the robot to operate for a suitable duration depending on the motor load and operating conditions.

5. LM2596 Voltage Regulator

The LM2596 is a DC-DC buck converter used to reduce the battery voltage to a suitable level for the electronic components. It has an adjustable output voltage and is used to provide a regulated supply to the ESP32 and other low-voltage circuits. This helps maintain stable operation of the control system.

6. Robot Chassis/Body

The robot chassis provides the mechanical structure for the Robot War. It supports and protects the ESP32, motor drivers, motors, battery, wheels, and other components. A strong and properly balanced chassis helps improve the stability and durability of the robot during movement.

7. Four Wheels

Four wheels are used to provide movement and support for the Robot War. The wheels are connected to the DC motors and convert the motor rotation into linear movement. Proper wheel arrangement provides better traction, balance, and directional control.

8. Relay Module

The relay module is used for switching electrical loads when required in the system. It allows the ESP32 control system to operate a higher-power electrical device using a low-power control signal. It can also provide electrical isolation between the control circuit and the connected load.

9. Jumper Wires

Jumper wires are used to establish electrical connections between the ESP32, BTS7960 motor drivers, relay module, voltage regulator, battery, and other components. They are useful for connecting both power and control signals during assembly, testing, and troubleshooting of the Robot War.

2.4 Feasibility Study

The feasibility study is carried out to determine whether the proposed ESP32-based Robot War can be successfully developed using the available hardware, software, technical knowledge, and resources. The project is evaluated based on technical, economic, operational, and time feasibility.

2.4.1 Technical Feasibility

The proposed system is technically feasible because the required components such as ESP32, 300 RPM DC motors, BTS7960 motor drivers, 12V 7Ah battery, and LM2596 voltage regulator are suitable for the intended application. The ESP32 provides the required control and wireless communication capabilities, while the BTS7960 drivers can control the DC motors. The system can be programmed using Arduino IDE and Embedded C/C++.

2.4.2 Economic Feasibility

The project is economically feasible because it uses commonly available and relatively low-cost electronic and mechanical components. The selected ESP32, motor drivers, DC motors, battery, and regulator provide the required functionality without requiring expensive industrial equipment. Therefore, the overall development cost can be maintained within a reasonable range.

2.4.3 Operational Feasibility

The proposed Robot War is easy to operate through wireless control. The ESP32 processes the received commands and controls the motors according to the required direction. The simple control mechanism allows the robot to be operated and tested conveniently while also providing scope for future improvements.

2.4.4 Time Feasibility

The project is feasible within the available development period because the hardware components and software tools are readily available. The development can be completed through systematic stages including component selection, circuit connections, programming, mechanical assembly, testing, and final integration.

Overall, the proposed Robot War is technically, economically, operationally, and time feasible. The selected components and development approach provide a practical foundation for implementing a functional ESP32-based Robot War prototype.

CHAPTER – 3

ARCHITECTURAL DESIGN

3.1 Modules Design

The proposed Robot War system is divided into several functional modules to simplify the design, implementation, and testing of the robot. Each module performs a specific function and works together with the other modules to achieve reliable robotic operation. The major modules include the ESP32 control module, wireless communication module, motor driver module, DC motor and movement module, power supply module, and mechanical structure module.

3.1.1 ESP32 Control Modul

The ESP32 acts as the main control unit of the Robot War. It receives the movement commands and processes them according to the programmed control logic. The ESP32 generates the required signals for the BTS7960 motor drivers to control the connected DC motors.

3.1.2 Wireless Communication Module

The wireless communication module is used to send movement commands to the Robot War. The ESP32 receives these commands and transfers them to the control program. This provides convenient wireless operation without requiring a physical control connection.

3.1.3 Motor Driver Module

The BTS7960 motor driver modules are used as an interface between the ESP32 and the high-current DC motors. They receive control signals from the ESP32 and provide the required current to operate the motors. The drivers support forward and reverse motor operation for directional movement.

3.1.4 DC Motor and Movement Module

The 300 RPM DC motors provide the mechanical driving force required to move the Robot War. The motors are connected to the wheels and are controlled through the BTS7960 drivers. By controlling the motor operation, the robot can perform forward, backward, left, and right movements.

3.1.5 Power Supply Module

The power supply module provides electrical power to the complete system. A 12V 7Ah battery is used as the main power source, while the LM2596 voltage regulator reduces the voltage to a suitable level for the ESP32 and other electronic components. Proper power distribution helps maintain stable system operation.

3.1.6 Mechanical Structure Module

The mechanical structure consists of the robot chassis, wheels, motors, battery, and electronic components. The components are arranged and mounted on the chassis to provide proper balance, strength, and stability. The mechanical structure supports the robot during movement and controlled Robot War operation.

3.1.7 System Integration and Testing Module

All the individual modules are integrated to form the complete Robot War system. The integrated system is tested for wireless response, motor operation, directional movement, speed, stability, and overall performance. Testing ensures that the hardware and software modules work together correctly.

3.2 Methodology & Algorithm Design

The methodology of the proposed Robot War is based on the integration of the ESP32 controller, wireless communication, BTS7960 motor drivers, DC motors, power supply, and mechanical structure. The ESP32 is programmed using Embedded C/C++ to receive and process movement commands. The processed commands are provided to the BTS7960 motor drivers to control the 300 RPM DC motors. The methodology includes controller initialization, wireless command processing, motor control, directional movement, power management, system integration, and performance testing. The complete system is tested for forward, backward, left, and right movement, speed, stability, motor performance, and wireless response.

3.2.1 ESP32 Control Algorithm

The ESP32 control algorithm is responsible for controlling the overall operation of the Robot War. Initially, the ESP32 initializes the required control pins and wireless communication. The controller then continuously waits for movement commands from the wireless control system. When a command is received, the ESP32 processes it and generates the corresponding signals for the

BTS7960 motor drivers. The drivers then operate the DC motors according to the selected movement. This process continues continuously to provide responsive and reliable control of the Robot War.

3.2.2 Wireless Command Processing

The wireless command processing method is used to provide remote control of the Robot War. Movement commands are transmitted from the control system to the ESP32 through wireless communication. The ESP32 receives the command, identifies the required operation, and passes the corresponding control signals to the motor drivers. This method allows the robot to be operated without a direct wired connection and provides convenient control during testing and operation.

3.2.3 Motor Driver Control

The BTS7960 motor driver is used as the interface between the ESP32 and the 300 RPM DC motors. Since the ESP32 cannot directly supply the current required by the motors, the BTS7960 drivers provide the required motor-driving capability. The ESP32 sends control signals to the drivers, and the drivers control the direction and operation of the connected motors. This arrangement provides reliable motor control for the Robot War.

3.2.4 Forward Movement

For forward movement, the ESP32 sends the appropriate control signals to the BTS7960 motor drivers. The connected DC motors rotate in the required direction, causing the wheels to move the Robot War forward. The movement continues while the forward command is active. Proper motor synchronization helps maintain stable and straight movement.

3.2.5 Backward Movement

For backward movement, the ESP32 provides the required reverse control signals to the BTS7960 motor drivers. The drivers operate the DC motors in the opposite direction, causing the wheels to move the Robot War backward. The controller continuously monitors the received command and maintains the required motor operation until another command is received.

3.2.6 Left Movement

For left movement, the ESP32 controls the motors so that the robot changes its direction toward the left side. The BTS7960 drivers provide the required control signals to the motors, producing the required difference in wheel movement. This allows the Robot War to turn and change its direction during operation.

3.2.7 Right Movement

For right movement, the ESP32 sends the corresponding control signals to the BTS7960 motor drivers. The motors operate in the required manner to turn the Robot War toward the right side. The controller continues to monitor the wireless commands so that the direction can be changed whenever required.

3.2.8 Power Management

The power management system provides the required electrical supply to the Robot War. A 12V 7Ah battery is used as the main power source for the robot. The LM2596 voltage regulator reduces the battery voltage to a suitable level for the ESP32 and other electronic components. Proper power distribution helps maintain stable operation of the motors and control electronics.

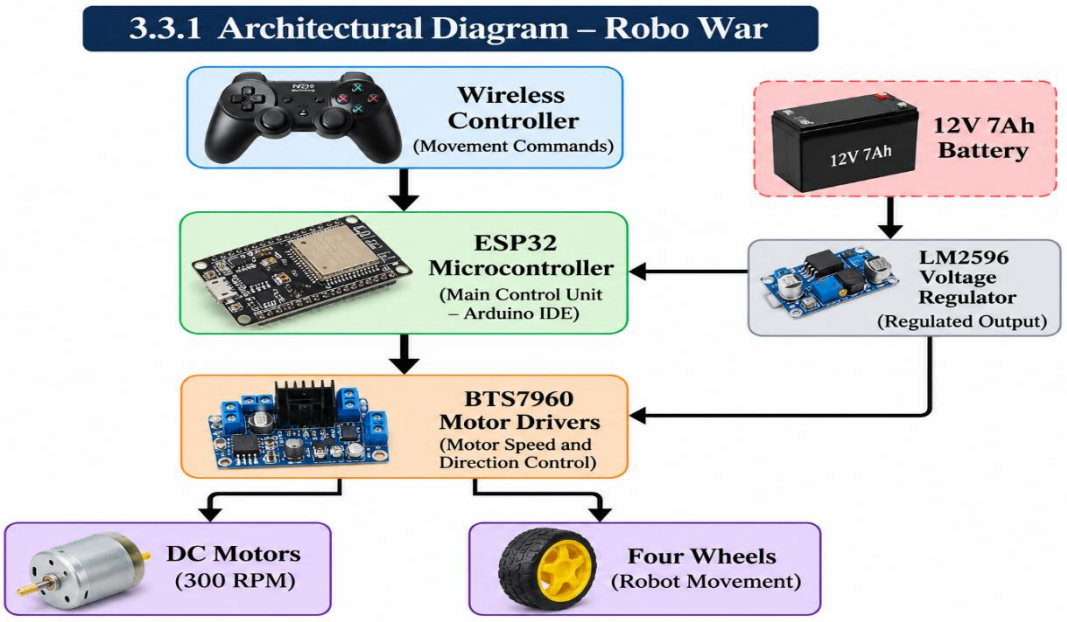
3.2.9 System Integration

System integration involves connecting the ESP32, BTS7960 motor drivers, DC motors, battery, LM2596 regulator, wheels, and chassis into a complete Robot War. The electrical and mechanical components are assembled according to the required design. The ESP32 control program is then tested with the connected hardware to verify correct communication and motor operation. Successful integration results in a complete working Robot War prototype.

3.2.10 Testing Methodology

The testing methodology is used to verify the performance and reliability of the completed Robot War. The robot is tested for forward, backward, left, and right movement along with motor performance, speed, stability, and wireless response. The power supply and electrical connections are also checked during operation. Any identified connection, programming, or movement problems are corrected and the system is tested again. Finally, the complete robot is evaluated under controlled operating conditions to confirm proper functionality.

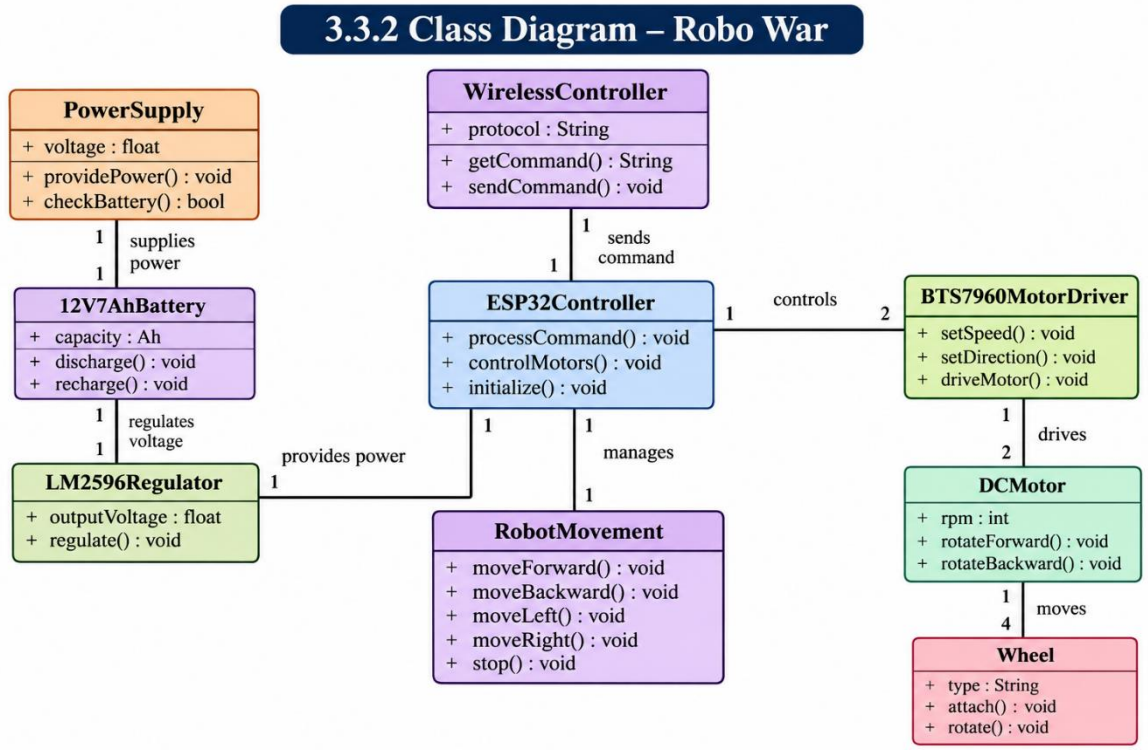
3.3 System Architecture



The architectural diagram shows the overall working structure of the Robot War. The ESP32 receives wireless commands and sends control signals to the BTS7960 motor drivers. The drivers operate the 300 RPM DC motors for forward, backward, left, and right movement. A 12V 7Ah battery supplies power, while the LM2596 provides regulated voltage for the electronic components.

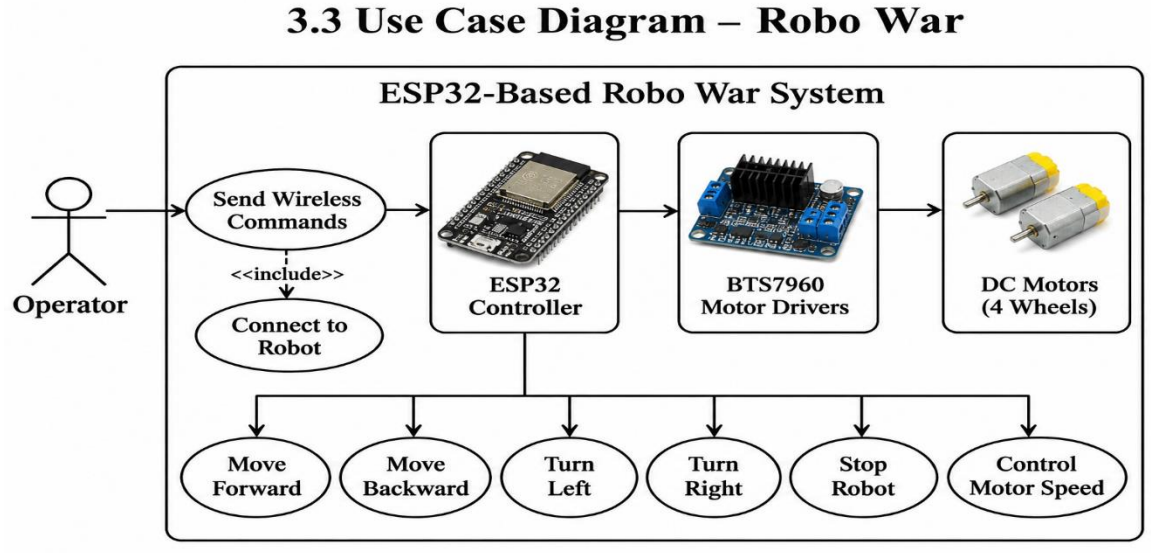
3.3.2 Class Diagram

The class diagram represents the main functional classes of the Robot War system and their relationships. The ESP32Controller manages the received wireless commands and controls the BTS7960 motor drivers. The motor drivers operate the DC motors to achieve the required robot movement. The PowerSupply and LM2596Regulator provide the required electrical power to the controller and motor system. This structure shows the interaction between the control, motor, movement, and power modules.



3.3.3 Use Case Diagram

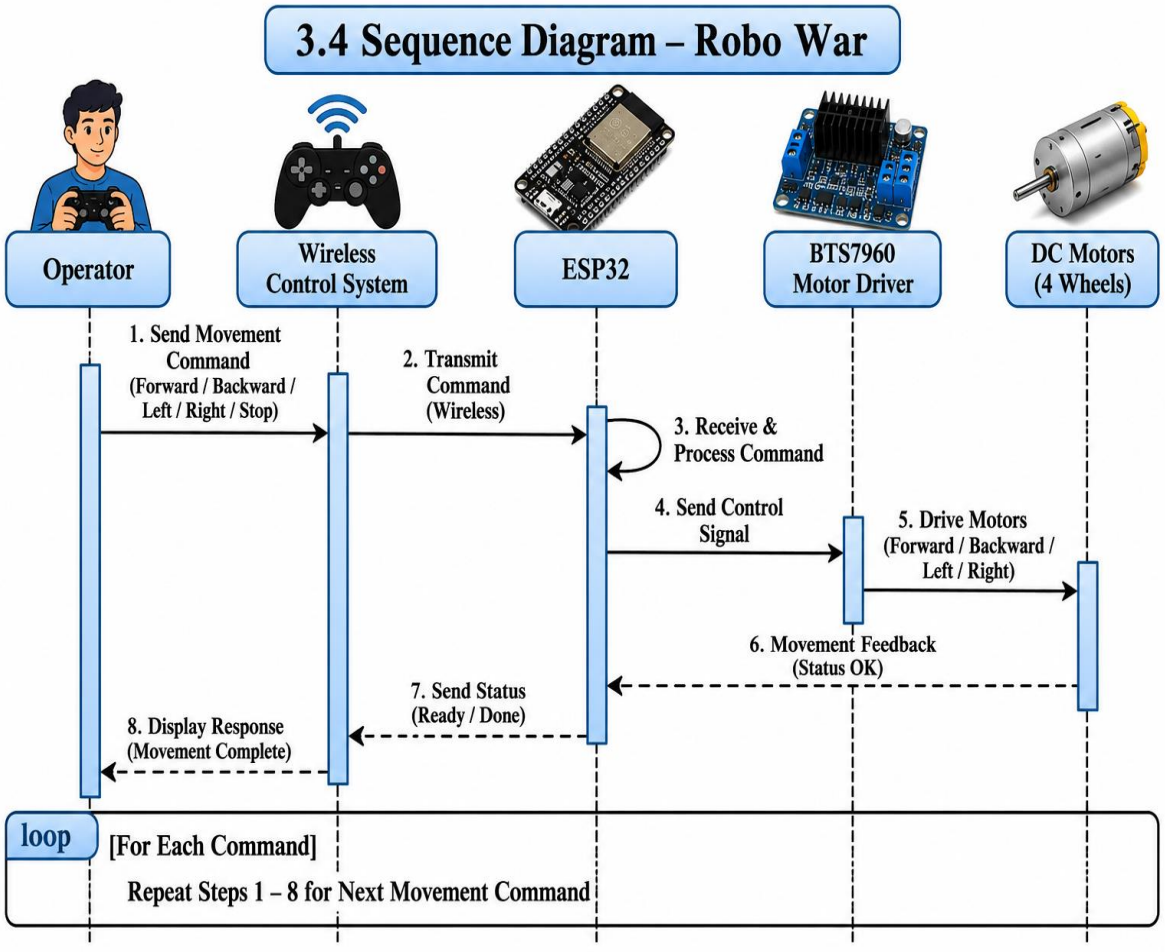
The use case diagram represents the interaction between the operator and the ESP32-based Robot War system. The operator sends wireless commands to control the robot and can perform forward, backward, left, right, and stop operations. The ESP32 processes the received commands and controls the motors through the BTS7960 motor drivers. This diagram shows the main functions available to the operator during Robot War operation.



3.3.4 Sequence Diagram

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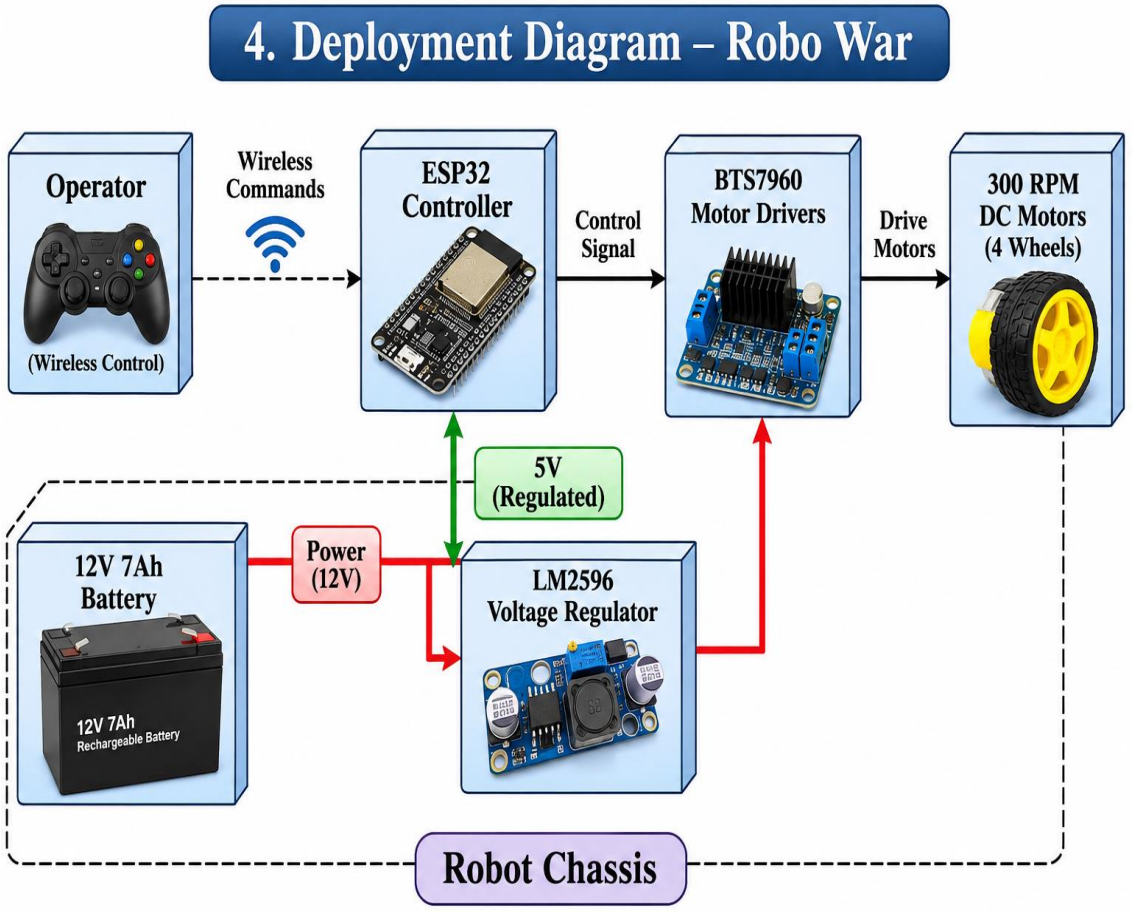
The sequence diagram represents the sequence of interactions between the operator, wireless control system, ESP32, BTS7960 motor driver, and DC motors. The operator first sends a movement command through the wireless control system. The ESP32 receives and processes the command and sends the required control signal to the BTS7960 motor driver. The motor driver then operates the DC motors according to the command, resulting in the required movement of the Robot War. This sequence continues for each movement command received from the operator



3.3.5 Deployment Diagram

The deployment diagram represents the physical arrangement of the Robot War hardware components. The operator sends wireless commands to the ESP32 controller, which communicates with the BTS7960 motor drivers. The motor drivers control the 300 RPM DC motors to produce the

required wheel movement. The 12V 7Ah battery supplies power to the motor drivers, while the LM2596 regulator provides suitable regulated voltage to the ESP32 and electronic components. The components are mounted on the robot chassis to form the complete Robot War system.



CHAPTER – 4

IMPLEMENTATION & TESTING

4.1 Coding Blocks

4.1.1 Main Coding Block

```
#include <BLEDevice.h>
#include <BLEServer.h>
#include <BLEUtils.h>
#include <BLE2902.h>

// BLE UUIDs
#define SERVICE_UUID           "6E400001-B5A3-F393-E0A9-E50E24DCCA9E"
#define CHARACTERISTIC_UUID_RX "6E400002-B5A3-F393-E0A9-E50E24DCCA9E"
#define CHARACTERISTIC_UUID_TX "6E400003-B5A3-F393-E0A9-E50E24DCCA9E"

BLEServer *pServer = NULL;
BLECharacteristic *pTxCharacteristic;
bool deviceConnected = false;

// Auxiliary relay
#define RELAY_PIN 23

// Left Motor
#define RPWM_L 25
#define LPWM_L 26
#define REN_L 27
#define LEN_L 14

// Right Motor
#define RPWM_R 32
#define LPWM_R 33
#define REN_R 18
#define LEN_R 19

int speedValue = 150;
```

```
String currentCommand = "stop";

void forward();
void backward();
void left();
void right();
void stopMotors();
void auxiliaryOn();
void auxiliaryOff();

class MyServerCallbacks : public BLEServerCallbacks {
  void onConnect(BLEServer* pServer) {
    deviceConnected = true;
    Serial.println("Bluetooth Client Connected");
  }

  void onDisconnect(BLEServer* pServer) {
    deviceConnected = false;
    stopMotors();
    auxiliaryOff();

    Serial.println("Bluetooth Client Disconnected");
    pServer->getAdvertising()->start();
  }
};

void setup() {
  Serial.begin(115200);

  pinMode(RPWM_L, OUTPUT);
  pinMode(LPWM_L, OUTPUT);
  pinMode(REN_L, OUTPUT);
  pinMode(LEN_L, OUTPUT);

  pinMode(RPWM_R, OUTPUT);
```

```
pinMode(LPWM_R, OUTPUT);
pinMode(REN_R, OUTPUT);
pinMode(LEN_R, OUTPUT);

pinMode(RELAY_PIN, OUTPUT);

digitalWrite(REN_L, HIGH);
digitalWrite(LEN_L, HIGH);
digitalWrite(REN_R, HIGH);
digitalWrite(LEN_R, HIGH);

stopMotors();
auxiliaryOff();

BLEDevice::init("ESP32_Robot");

pServer = BLEDevice::createServer();
pServer->setCallbacks(new MyServerCallbacks());

BLEService *pService =
   pServer->createService(SERVICE_UUID);

pTxCharacteristic = pService->createCharacteristic(
   CHARACTERISTIC_UUID_TX,
   BLECharacteristic::PROPERTY_NOTIFY);

pTxCharacteristic->addDescriptor(new BLE2902());

pService->start();

BLEAdvertising *pAdvertising =
   BLEDevice::getAdvertising();

pAdvertising->addServiceUUID(SERVICE_UUID);
pAdvertising->setScanResponse(true);
```

```
  BLEDevice::startAdvertising();

  Serial.println("BLE Ready");
}

void loop() {
  delay(20);
}
```

4.1.2 Functional Code Block 1

```
class MyCallbacks : public BLECharacteristicCallbacks {

  void onWrite(BLECharacteristic *pCharacteristic) {

    String rxValue =
       pCharacteristic->getValue().c_str();

    if (rxValue.length() > 0) {

      rxValue.trim();

      Serial.print("Received Command: ");
      Serial.println(rxValue);

      if (rxValue.startsWith("speed:")) {

        speedValue =
           rxValue.substring(6).toInt();

        speedValue =
           constrain(speedValue, 0, 255);
      }

      else {
```

```
      currentCommand = rxValue;

      if (currentCommand == "forward")
        forward();

      else if (currentCommand == "backward")
        backward();

      else if (currentCommand == "left")
        left();

      else if (currentCommand == "right")
        right();

      else if (currentCommand == "stop")
        stopMotors();

      else if (currentCommand == "aux_on")
        auxiliaryOn();

      else if (currentCommand == "aux_off")
        auxiliaryOff();
    }
   }
  }
};
```

4.1.3 Functional Code Block 2

```
void forward() {

  analogWrite(RPWM_L, speedValue);
  analogWrite(LPWM_L, 0);

  analogWrite(RPWM_R, speedValue);
  analogWrite(LPWM_R, 0);
}


void backward() {

  analogWrite(RPWM_L, 0);
  analogWrite(LPWM_L, speedValue);

  analogWrite(RPWM_R, 0);
  analogWrite(LPWM_R, speedValue);
}


void left() {

  analogWrite(RPWM_L, 0);
  analogWrite(LPWM_L, speedValue);

  analogWrite(RPWM_R, speedValue);
  analogWrite(LPWM_R, 0);
}


void right() {

  analogWrite(RPWM_L, speedValue);
  analogWrite(LPWM_L, 0);
```

```
  analogWrite(RPWM_R, 0);
  analogWrite(LPWM_R, speedValue);
}


void stopMotors() {

  analogWrite(RPWM_L, 0);
  analogWrite(LPWM_L, 0);

  analogWrite(RPWM_R, 0);
  analogWrite(LPWM_R, 0);
}


void auxiliaryOn() {

  digitalWrite(RELAY_PIN, HIGH);

  Serial.println("Auxiliary Relay ON");
}


void auxiliaryOff() {

  digitalWrite(RELAY_PIN, LOW);

  Serial.println("Auxiliary Relay OFF");
}

Endmodule
```

4.1.4 Code Execution Flow Chart

The code execution flow chart represents the sequence of operations performed by the ESP32 Robot War program. The ESP32 first initializes the motor pins, relay, BLE communication, and required variables. After initialization, the system starts BLE advertising and waits for a command from the operator. When a command is received, the ESP32 identifies the command and performs the corresponding motor or auxiliary relay operation. The system continuously waits for new commands until the Bluetooth connection is disconnected.

4.2 Test Case Study

The Robot War was tested to verify the proper operation of the ESP32 controller, BLE communication, motor control, speed control, stopping function, and auxiliary relay. Different commands were sent through the Bluetooth control interface, and the corresponding robot response was observed. The robot was tested for forward, backward, left, and right movement, along with stop and speed-control commands. The auxiliary relay was also tested for ON and OFF operation. The Bluetooth disconnection condition was checked to ensure that the motors stop and the auxiliary relay turns OFF. The test results confirmed that the developed Robot War performed the required operations correctly.

4.2.1 BLE Communication Test

The BLE communication test was conducted to verify the wireless connection between the operator control interface and the ESP32. The ESP32 was configured as a BLE device and made available for connection through the control interface. After establishing the connection, different movement commands were transmitted to the controller. The ESP32 successfully received the commands and displayed the received information through the Serial Monitor. The connection was also tested during disconnection to verify that the system responds safely. The test confirmed that BLE communication provides a suitable method for sending control commands to the Robot War**.**

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4.2.2 Motor Movement Test

The motor movement test was performed to verify the operation of the BTS7960 motor drivers and

300 RPM DC motors. Forward, backward, left, and right commands were sent through the BLE control interface. For each command, the ESP32 processed the received instruction and generated the required motor control signals. The motors responded according to the selected direction, and the stop command successfully stopped the motors. Speed values were also tested by sending different speed commands within the programmed range. The test confirmed that the motor control system provides the required directional movement and speed control.

4.2.3 Auxiliary Relay Test

The auxiliary relay test was conducted to verify manual ON and OFF control through the ESP32. The aux_on command was transmitted through the BLE control interface, and the relay was activated by the ESP32. The aux_off command was then sent to deactivate the relay. The relay response was observed for both commands to verify correct operation. The system was also checked during Bluetooth disconnection, where the auxiliary relay was programmed to return to the OFF state. The test confirmed that the auxiliary relay can be controlled reliably through the wireless interface.

CHAPTER – 5

RESULTS AND DISCUSSION

5.1 Resulting Images

5.1.1 Screen shot 1

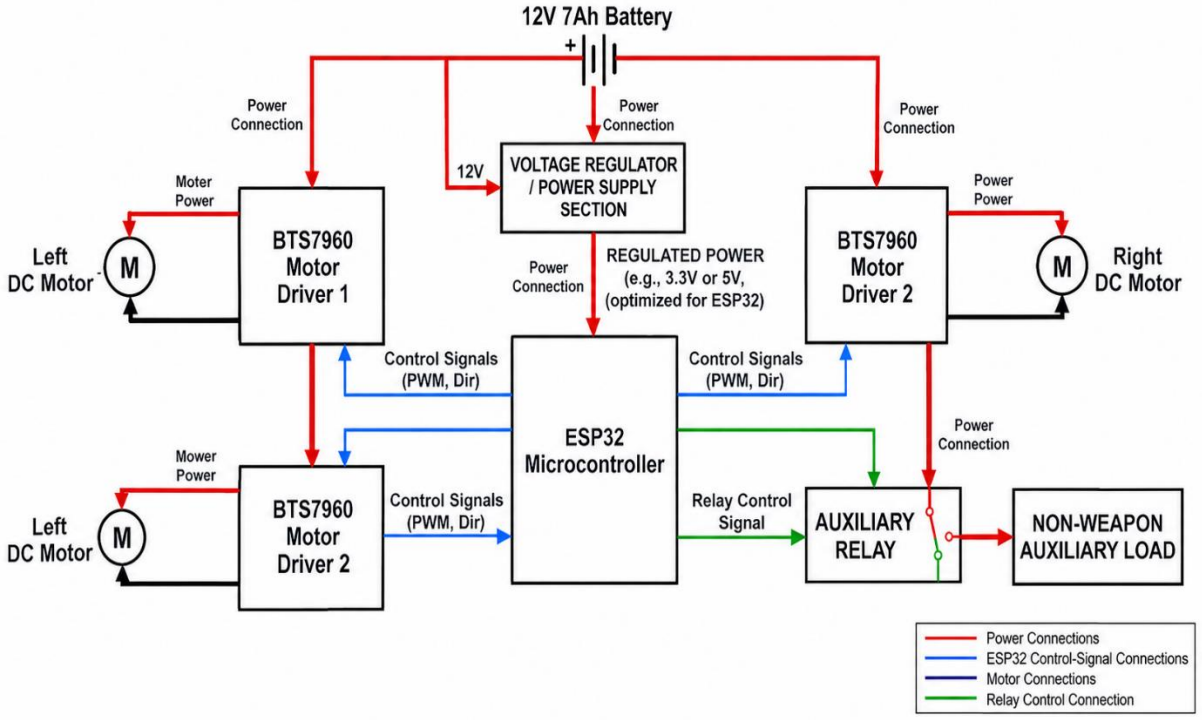
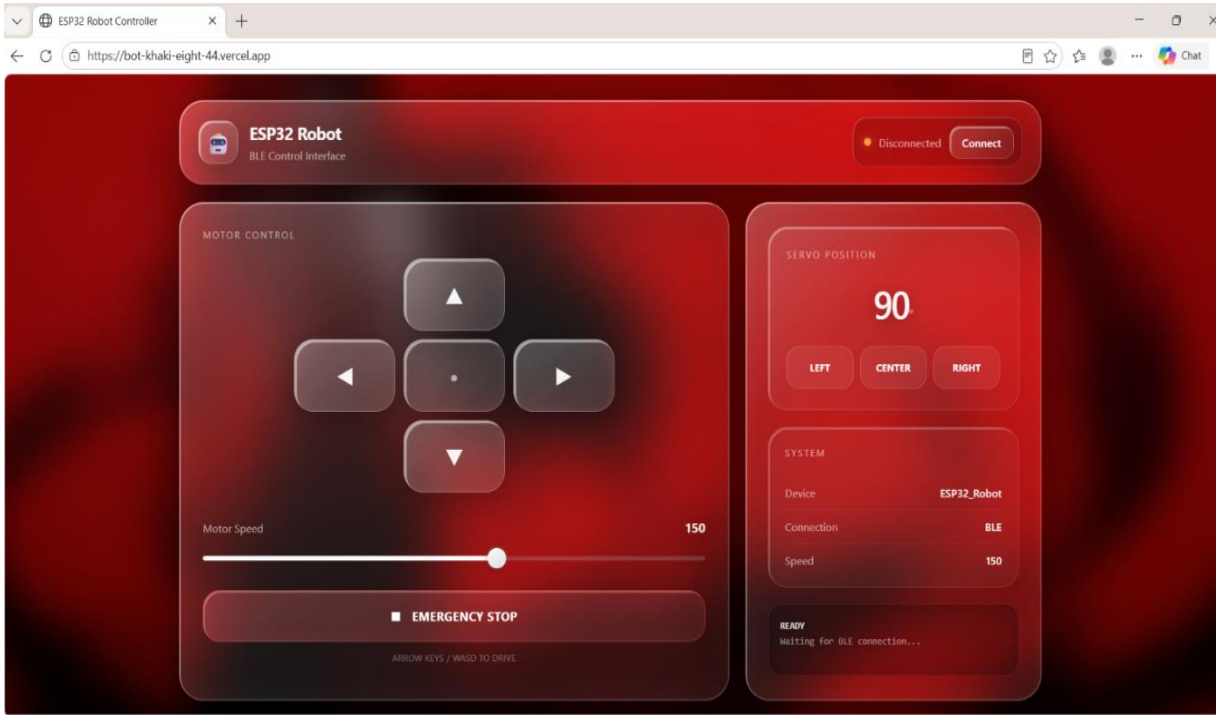


Figure 4.X: Circuit Diagram of **Robo War**

Fig 5.1

The circuit diagram shows the electrical connections of the ESP32-based Robot War. A 12V 7Ah battery is used as the main power source and supplies power to the two BTS7960 motor drivers. The left and right DC motors are connected to the respective motor drivers, while the ESP32 provides the required control signals to operate them. An LM2596 voltage regulator is used to provide a suitable regulated supply to the ESP32 and other electronic components

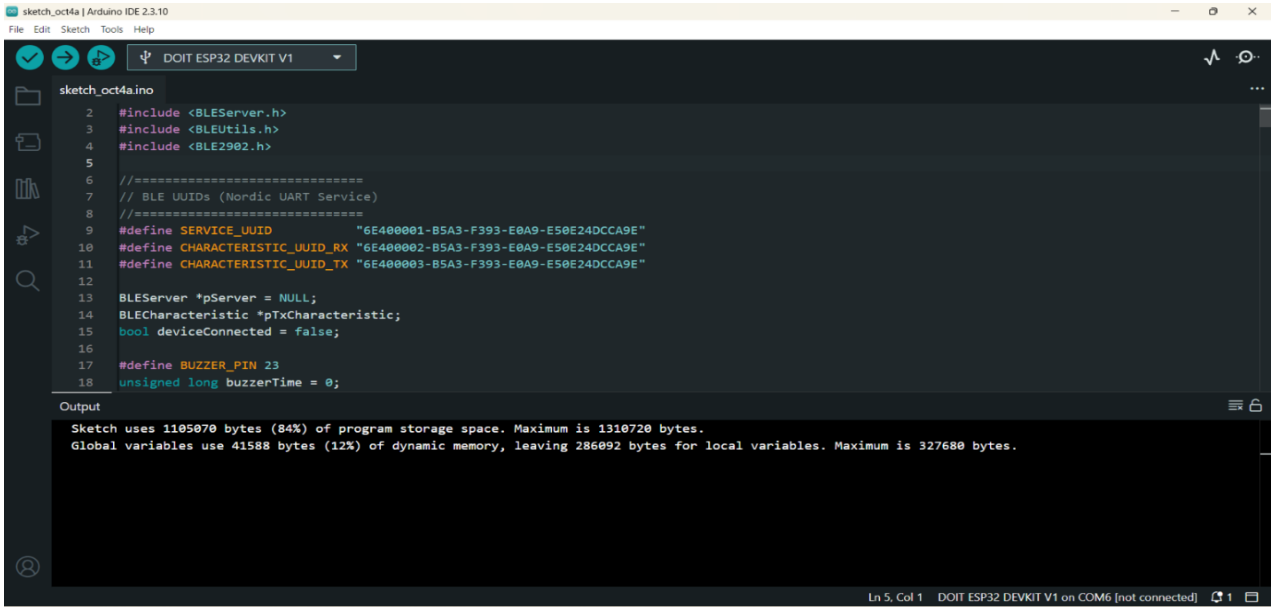
5.1.2 Screen shot 2



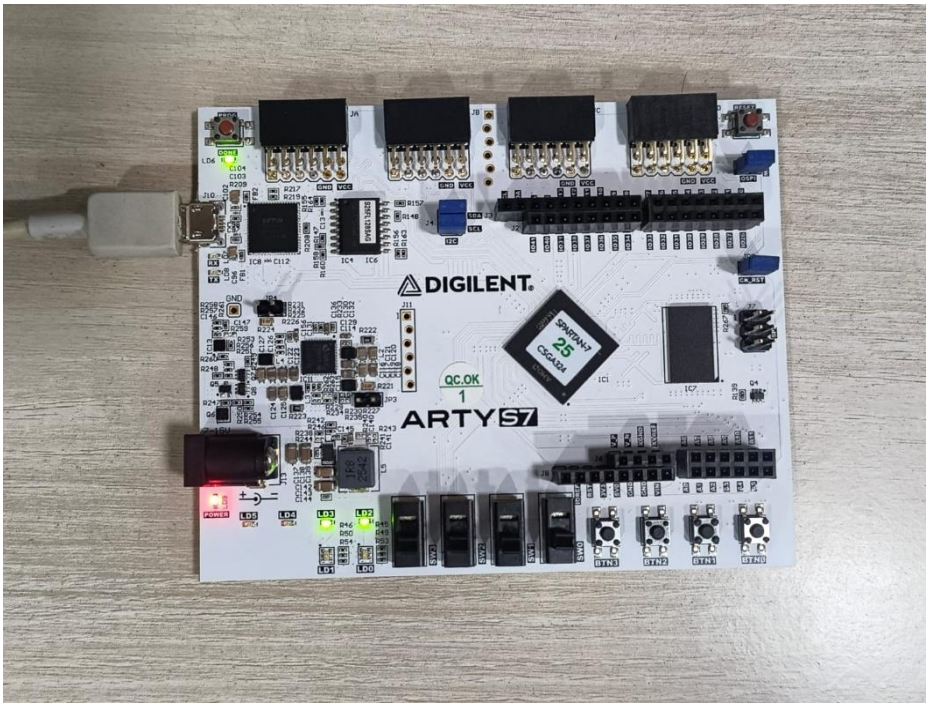
ESP32 Robot Controller Interface

The developed web interface provides a simple and user-friendly control system for the ESP32-based Robot War. The interface includes a Bluetooth connection status and a Connect button for establishing communication with the ESP32 robot. The Motor Control section provides directional buttons for forward, backward, left, right, and stop operations. A motor-speed slider is provided to adjust the required speed during operation. The interface also includes an Emergency Stop button for quickly stopping the robot. A system information section displays the device name, connection type, and current speed. The graphical interface provides convenient wireless control and real-time operating information for the Robot War.

5.1.3 Screen shot 3



The ESP32 Robot War program was developed and compiled using the Arduino IDE. The program includes the required BLE libraries and Nordic UART Service configuration for wireless communication. The ESP32 is programmed to receive control commands and operate the motor drivers according to the selected movement. The Arduino IDE also provides compilation and memory-usage information, which helps verify that the program fits within the ESP32 resources. The compiled program is uploaded to the ESP32 board through the selected COM port for hardware testing and operation.



5.2 Resulting Tables

5.2.1 Table 1

5.2.1 Table 1 – Hardware Components and Specifications

S.No.	Component	Specification	Quantity
1	 **ESP32 Microcontroller**	Wi-Fi/Bluetooth enabled controller	1
2	 **DC Motor**	300 RPM	5
3	 **BTS7960 Motor Driver**	High-current DC motor driver	2
4	 **Battery**	12V, 7Ah	1
5	 **LM2596 Voltage Regulator**	Adjustable DC-DC Buck Converter	1
6	 **Wheels**	Robot wheels	4
7	 **Relay Module**	Electrically controlled switching module	1
8	 **Robot Chassis**	Mechanical robot body	1
9	 **Jumper Wires**	Electrical connection wires	As required

Table 5.2.1 presents the major hardware components used in the development of the ESP32-based Robot War. The system uses an ESP32 microcontroller as the main control unit, 300 RPM DC motors for movement, and BTS7960 motor drivers for controlling the motors. A 12V 7Ah battery provides the main power supply, while an LM2596 voltage regulator provides regulated voltage to the electronic components. Four wheels and a robot chassis provide the required mechanical support and movement.

5.2.2 Table 2

Static Timing Analysis Metrics

S.No.	Parameter	Specification / Value	Purpose
1	Main Battery	12V, 7Ah	Main power source
2	Battery Energy	84 Wh (nominal)	Stored energy
3	Motor Supply	12V	DC motor power
4	Voltage Regulator	LM2596 DC-DC Buck Converter	Regulated supply
5	Controller Supply	Regulated DC Supply	ESP32 power
6	Motor Driver	BTS7960 × 2	Motor control
7	Motor Speed	300 RPM	Robot movement
8	Power Efficiency	Load dependent	Depends on operating conditions
9	Operating Stability	Tested during movement	System performance

Table 5.2.2: Power and Efficiency Parameters of the ESP32-Based Battle Bot

Table 5.2.2 presents the power supply and efficiency parameters of the developed ESP32-based Robot War. The robot uses a 12V, 7Ah battery as the main power source, providing a nominal stored energy of approximately 84 Wh. The battery supplies power to the BTS7960 motor drivers, which control the 300 RPM DC motors used for robot movement. An LM2596 DC-DC buck converter is used to provide a suitable regulated voltage for the ESP32 and other electronic components. The overall efficiency of the system depends on factors such as motor load, battery condition, wiring losses, and voltage regulator performance. The power system was tested during robot movement to verify stable operation of the motors and electronic components.

5.3 Results Analysis

The developed ESP32-based Robot War was tested to evaluate its overall performance under different operating conditions. The analysis focused on the power requirements, timing response, motor movement, wireless communication, and stability of the robot. The test results were used to verify whether the developed system performed according to the intended project objectives. The ESP32 successfully processed the received control commands and operated the BTS7960 motor drivers for directional movement. The power supply and voltage regulation system also provided the required electrical power for the controller and motor system. The following subsections present the major results obtained from the developed Robot War.

5.3.1 Power Analysis

Figure 5.3.1 presents the power analysis of the ESP32-based Robot War. The robot uses a 12V, 7Ah battery as the main power source, providing a nominal stored energy of approximately 84 Wh. The battery supplies power to the BTS7960 motor drivers and the 300 RPM DC motors, while the LM2596 voltage regulator provides a suitable regulated supply for the ESP32 and other electronic components. The power requirements depend on the motor load, battery condition, and operating conditions. The analysis confirms that the selected power system is suitable for supporting the major electrical requirements of the Robot War during operation.

5.3.2 Timing Analysis

Figure 5.3.2 presents the timing analysis of the Robot War during wireless control operation. The response time represents the time required for the system to receive a command, process it through the ESP32, and generate the corresponding motor control signal. Forward, backward, left, right, and stop commands were considered during testing. The timing analysis helps evaluate the responsiveness of the control system and confirms the ability of the ESP32 to process movement commands effectively.

5.3.3 Motor Performance Analysis

The motor performance analysis was carried out to verify the operation of the 300 RPM DC motors and BTS7960 motor drivers. The robot was tested using forward, backward, left, right, and stop commands. The motors responded according to the received commands and provided the required directional movement. The test also verified that the motor drivers could control the motors according to the programmed speed and direction requirements. The results indicate that the motor

control system performs the required movement functions effectively.

5.3.4 Wireless Communication Analysis

The wireless communication system was tested to verify reliable command transmission between the operator control interface and the ESP32. Different movement and speed commands were transmitted to the robot during testing. The ESP32 successfully received and processed the commands and generated the corresponding control signals. The communication test demonstrated that the wireless control system provides convenient operation of the Robot War.

5.3.5 Overall Performance Analysis

The overall performance analysis was conducted by combining the results obtained from power, timing, motor, and wireless communication tests. The Robot War successfully performed the required directional movements and responded to the control commands. The ESP32 provided the required processing and control, while the BTS7960 drivers controlled the DC motors effectively. The power supply and voltage regulation system supported the operation of the electronic and motor components. Overall, the developed system achieved the main objectives of the project and demonstrated the practical operation of an ESP32-based Robot War prototype.

CHAPTER – 6
CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The ESP32-based Robot War was successfully designed and developed as a compact, reliable, and easily controllable robotic system. The project integrated an ESP32 microcontroller, 300 RPM DC motors, BTS7960 motor drivers, a 12V 7Ah battery, LM2596 voltage regulator, wheels, chassis, and wireless communication system into a single functional prototype. The ESP32 successfully processed the control commands and generated the required signals for controlling the motor drivers and achieving forward, backward, left, and right movement.

The developed system was tested for motor operation, wireless communication, power requirements, timing response, and overall movement performance. The BTS7960 motor drivers provided effective control of the DC motors, while the power supply system supported the operation of both the motors and electronic components. The testing results demonstrated that the Robot War could respond to the required control commands and perform the intended movements under controlled operating conditions.

Overall, the project provided practical experience in embedded programming, wireless communication, DC motor control, power management, hardware integration, and robotic system development. The successful development and testing of the prototype demonstrate the feasibility of using an ESP32-based control system for a Robot War. The developed system also provides a suitable foundation for future improvements and advanced robotic applications.

6.2 Future Scope

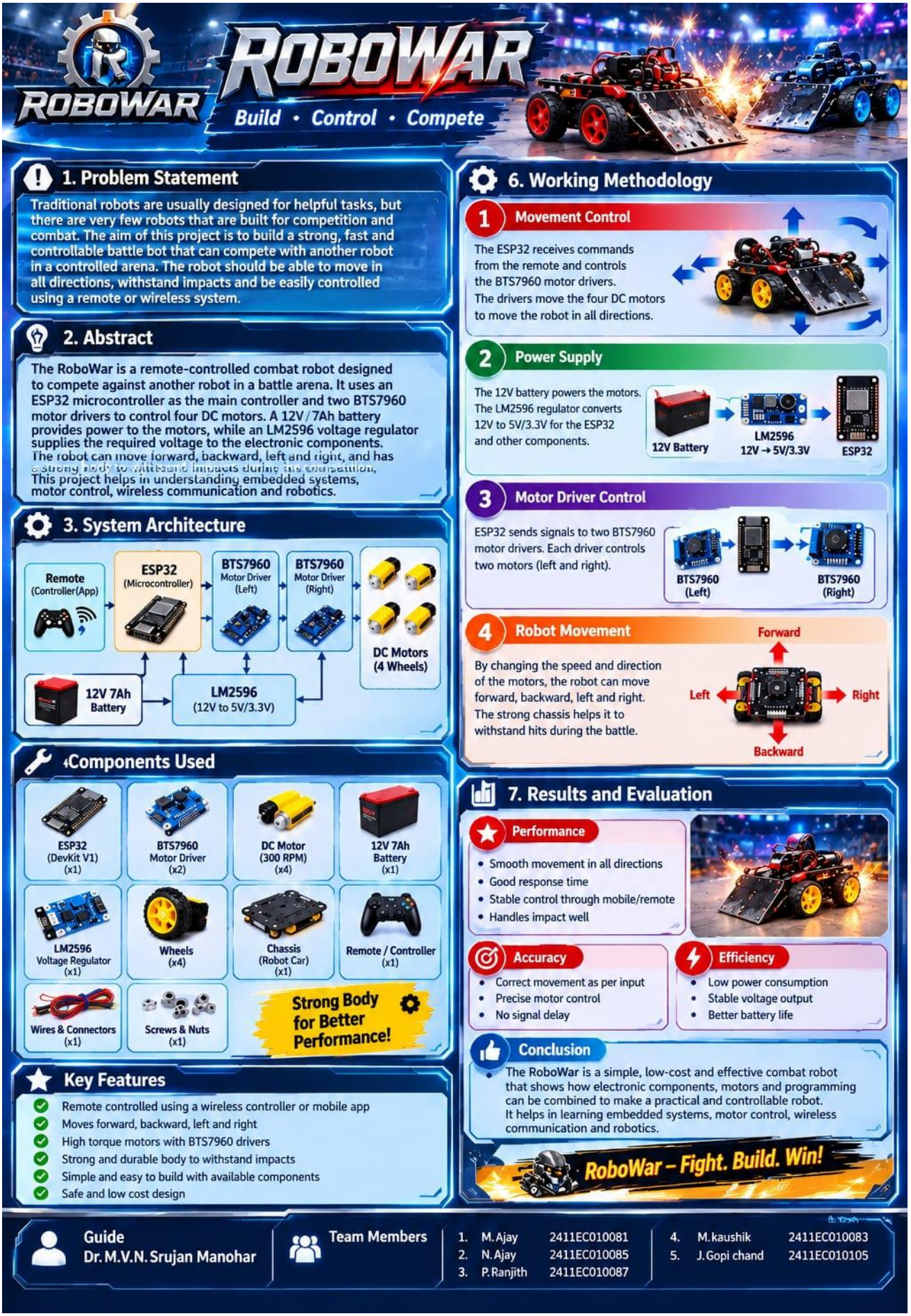
The developed ESP32-based Robot War provides a good foundation for further development in robotics and embedded systems. In future, the robot can be improved by using higher-torque and high-performance motors to achieve better movement and driving capability. The mechanical chassis can also be strengthened using lightweight and durable materials to improve stability and reliability during operation.

Additional sensors such as ultrasonic, infrared, and distance sensors can be integrated to provide better awareness of the surrounding environment. Battery monitoring and advanced power-management techniques can also be implemented to monitor voltage, current consumption, and battery condition. The wireless control system can be enhanced through a mobile or web-based interface for convenient real-time monitoring and operation.

Furthermore, encoder-based motor control can be added for accurate speed and movement control. Autonomous navigation and intelligent decision-making can also be considered for future versions. These improvements can make the Robot War more efficient, stable, powerful, and suitable for advanced robotic applications and future experimental developments.

References

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- **Espressif Systems, "ESP32 Technical Documents," Espressif Systems. Espressif Technical Documents**
- **Espressif Systems, "Arduino Core for ESP32," GitHub Repository. Arduino-ESP32 GitHub Repository**
- **Espressif Systems, "ESP32 Hardware and Peripheral Documentation," Espressif Systems. ESP32 Datasheet and Hardware Documentation**
- **These references support the ESP32 hardware, GPIO, Bluetooth/BLE, PWM/motor-control capabilities, power management, and Arduino IDE/ESP32 programming used in your Robot War project. (documentation.espressif.com)**


ROBOWAR
ROBOWAR
Build • Control • Compete
1. Problem Statement
Traditional robots are usually designed for helpful tasks, but there are very few robots that are built for competition and combat. The aim of this project is to build a strong, fast and controllable battle bot that can compete with another robot in a controlled arena. The robot should be able to move in all directions, withstand impacts and be easily controlled using a remote or wireless system.
2. Abstract
The RoboWar is a remote-controlled combat robot designed to compete against another robot in a battle arena. It uses an ESP32 microcontroller as the main controller and two BTS7960 motor drivers to control four DC motors. A 12V / 7Ah battery provides power to the motors, while an LM2596 voltage regulator supplies the required voltage to the electronic components. The robot can move forward, backward, left and right, and has a strong body to withstand impacts during the competition. This project helps in understanding embedded systems, motor control, wireless communication and robotics.
3. System Architecture
Remote (Controller/App)
ESP32 (Microcontroller)
BTS7960 Motor Driver (Left)
BTS7960 Motor Driver (Right)
DC Motors (4 Wheels)
12V 7Ah Battery
LM2596 (12V to 5V/3.3V)
Components Used
ESP32 (DevKit V1) (x1)
BTS7960 Motor Driver (x2)
DC Motor (300 RPM) (x4)
12V 7Ah Battery (x1)
LM2596 Voltage Regulator (x1)
Wheels (x4)
Chassis (Robot Car) (x1)
Remote / Controller (x1)
Wires & Connectors (x1)
Screws & Nuts (x1)
Strong Body for Better Performance!
Key Features
Remote controlled using a wireless controller or mobile app
Moves forward, backward, left and right
High torque motors with BTS7960 drivers
Strong and durable body to withstand impacts
Simple and easy to build with available components
Safe and low cost design
6. Working Methodology
1 Movement Control
The ESP32 receives commands from the remote and controls the BTS7960 motor drivers. The drivers move the four DC motors to move the robot in all directions.
2 Power Supply
The 12V battery powers the motors. The LM2596 regulator converts 12V to 5V/3.3V for the ESP32 and other components.
12V Battery
LM2596 12V → 5V/3.3V
ESP32
3 Motor Driver Control
ESP32 sends signals to two BTS7960 motor drivers. Each driver controls two motors (left and right).
BTS7960 (Left)
BTS7960 (Right)
4 Robot Movement
By changing the speed and direction of the motors, the robot can move forward, backward, left and right. The strong chassis helps it to withstand hits during the battle.
Forward
Left
Right
Backward
7. Results and Evaluation
Performance
Smooth movement in all directions
Good response time
Stable control through mobile/remote
Handles impact well
Accuracy
Correct movement as per input
Precise motor control
No signal delay
Efficiency
Low power consumption
Stable voltage output
Better battery life
Conclusion
The RoboWar is a simple, low-cost and effective combat robot that shows how electronic components, motors and programming can be combined to make a practical and controllable robot. It helps in learning embedded systems, motor control, wireless communication and robotics.
RoboWar – Fight. Build. Win!
Guide
Dr. M.V.N. Srujan Manohar
Team Members
1. M. Ajay 2411EC010081
2. N. Ajay 2411EC010085
3. P. Ranjith 2411EC010087
4. M. kaushik 2411EC010083
5. J. Gopi chand 2411EC010105